

Question:

Carlos is planting a lemon tree. The tree will cost \$90 to plant. Each year it will grow 7 lemons, which he can sell for \$1.5 each. It costs \$3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think> To determine how many years it will take before Carlos starts earning money from his lemon tree , we need to compare the annual cost with the annual income .

The initial cost to plant the lemon tree is \$ 9 0 . Each year , the tree will produce 7 le mons , each lemon can be sold for \$ 1 . 5 , so the annual income is $(7 \times 1 . 5 = 1 0 . 5)$ dollars .

The annual cost to water and feed the tree is \$ 3 .

To find out when Carlos starts earning money , we need to find the point where the annual income exceeds the annual cost .

Let (n) be the number of years . The total cost after (n) years is $(9 0 + 3 n)$ dollars . The total income after (n) years is $(1 0 . 5 n)$ dollars .

We set up the inequality :

$$[1 0 . 5 n > 9 0 + 3 n]$$

Sub tract $(3 n)$ from both sides :

$$[7 . 5 n > 9 0]$$

Divide both sides by 7 . 5 :

$$[n > \frac { 9 0 } { 7 . 5 }]$$

Calculate the right side :

$$[n > 1 2]$$

Since (n) must be an integer , we round up 1 2 to the next whole number , which is 1 3 .

Thus , it will take 1 3 years for Carlos to start earning money on the lemon tree . </think>

< answer > 1 3 </answer>